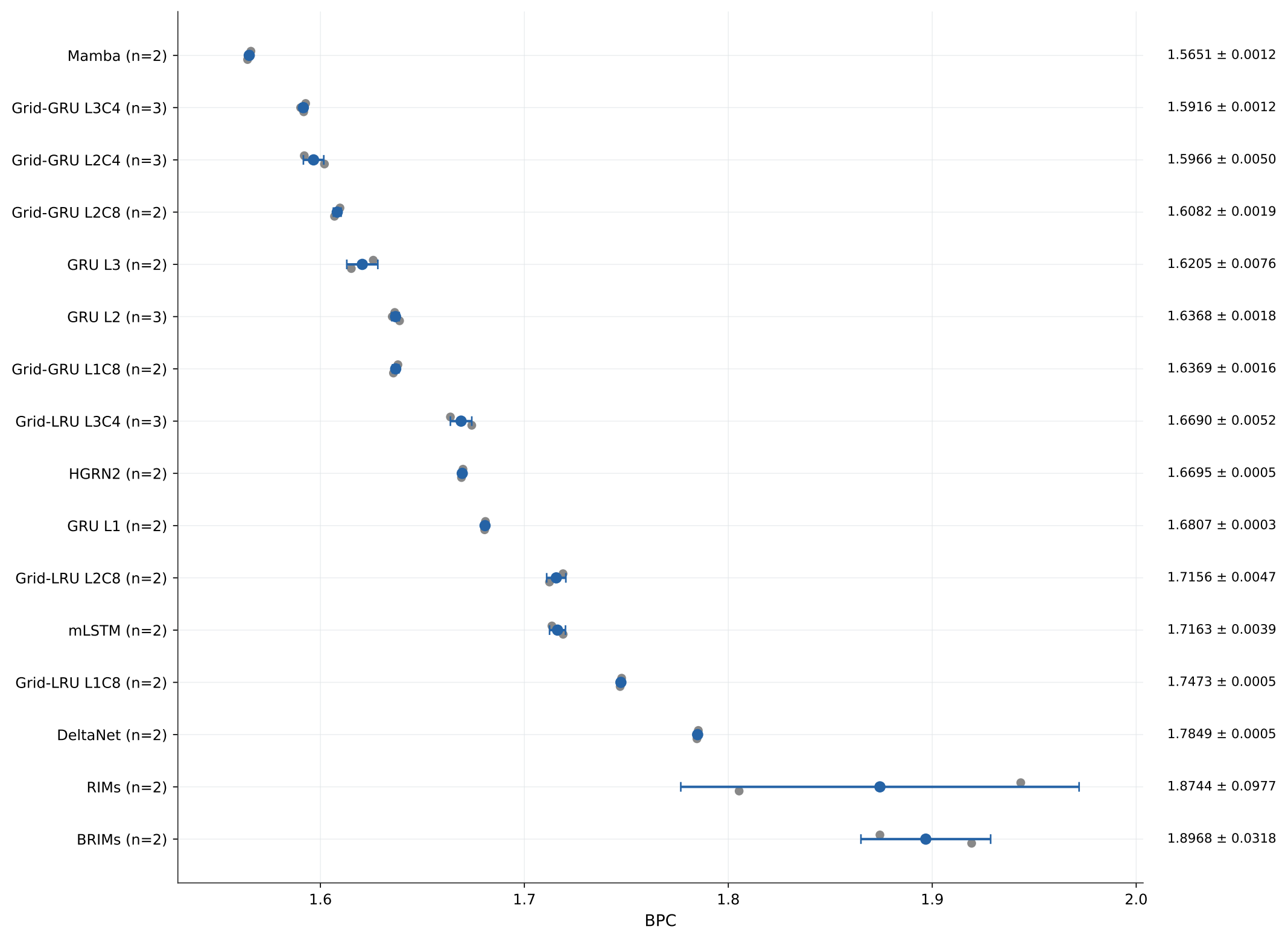
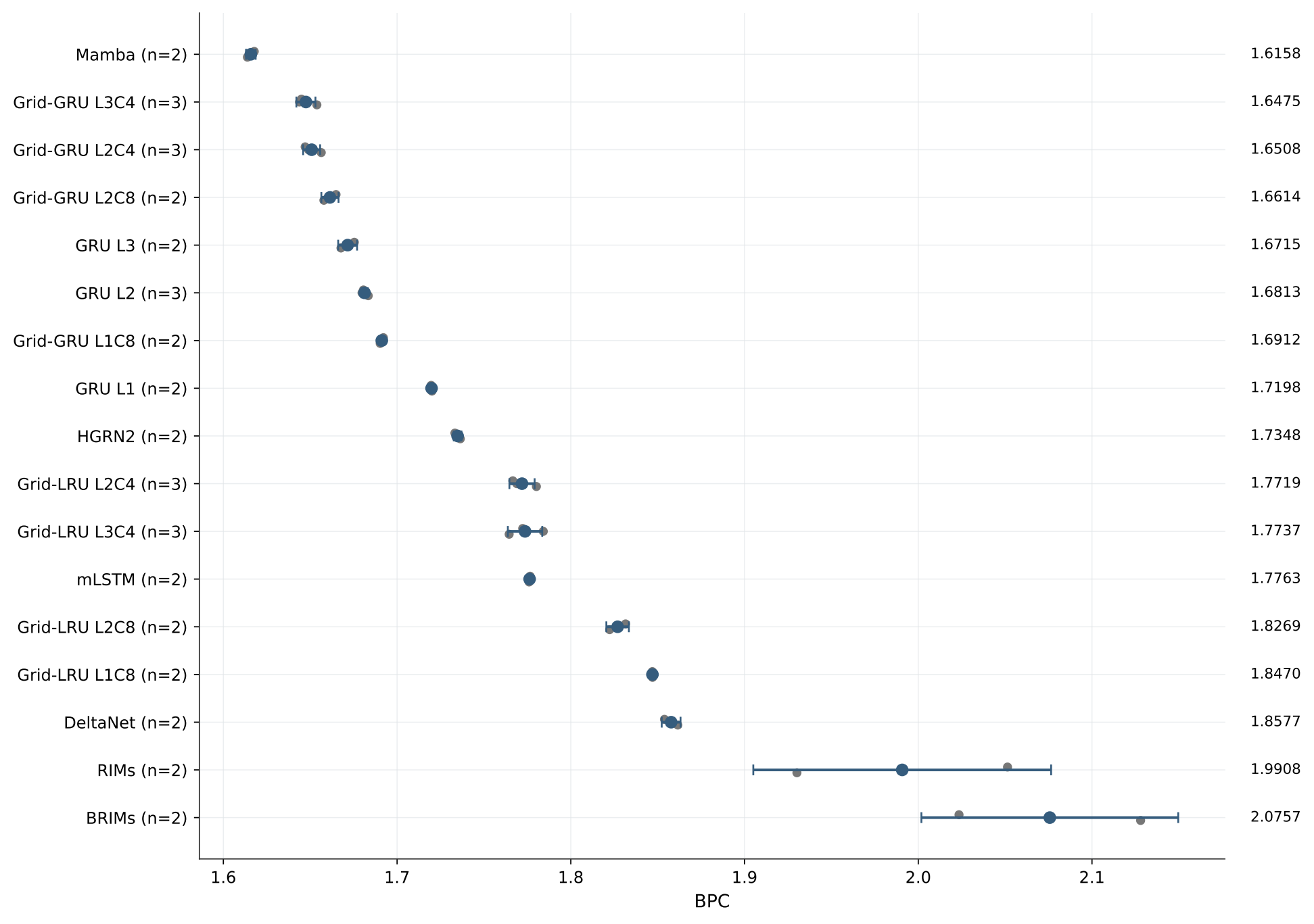
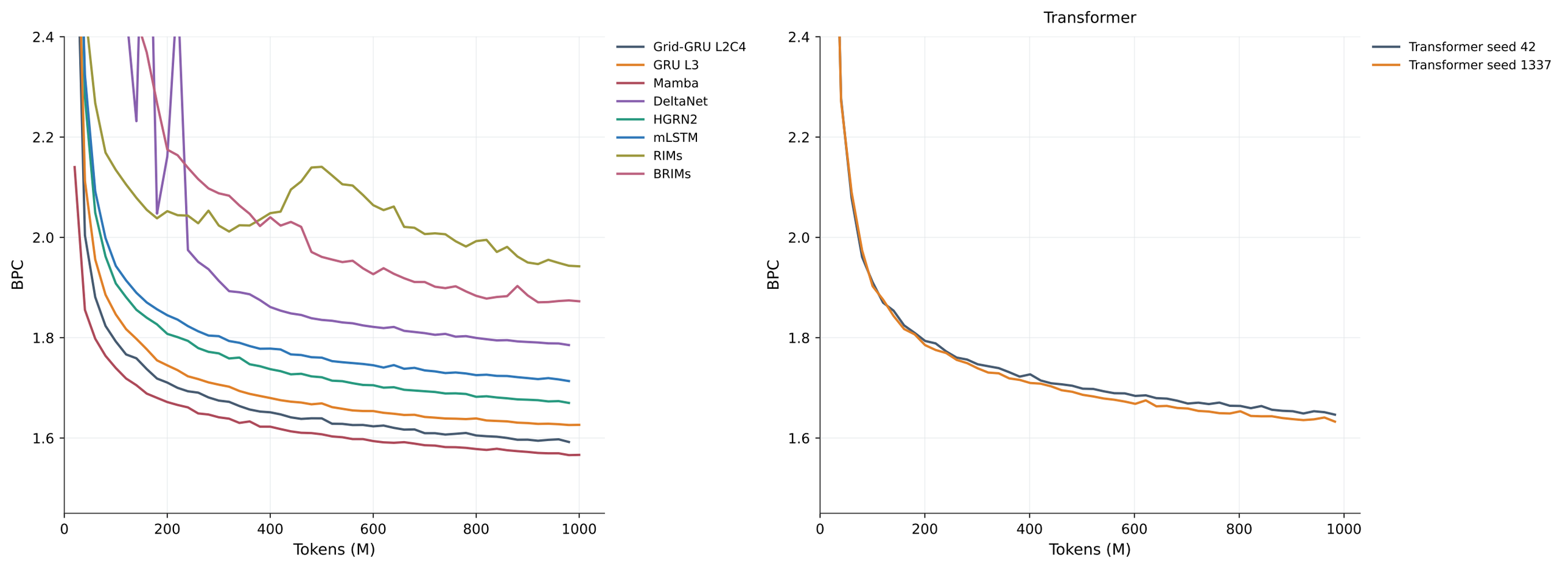
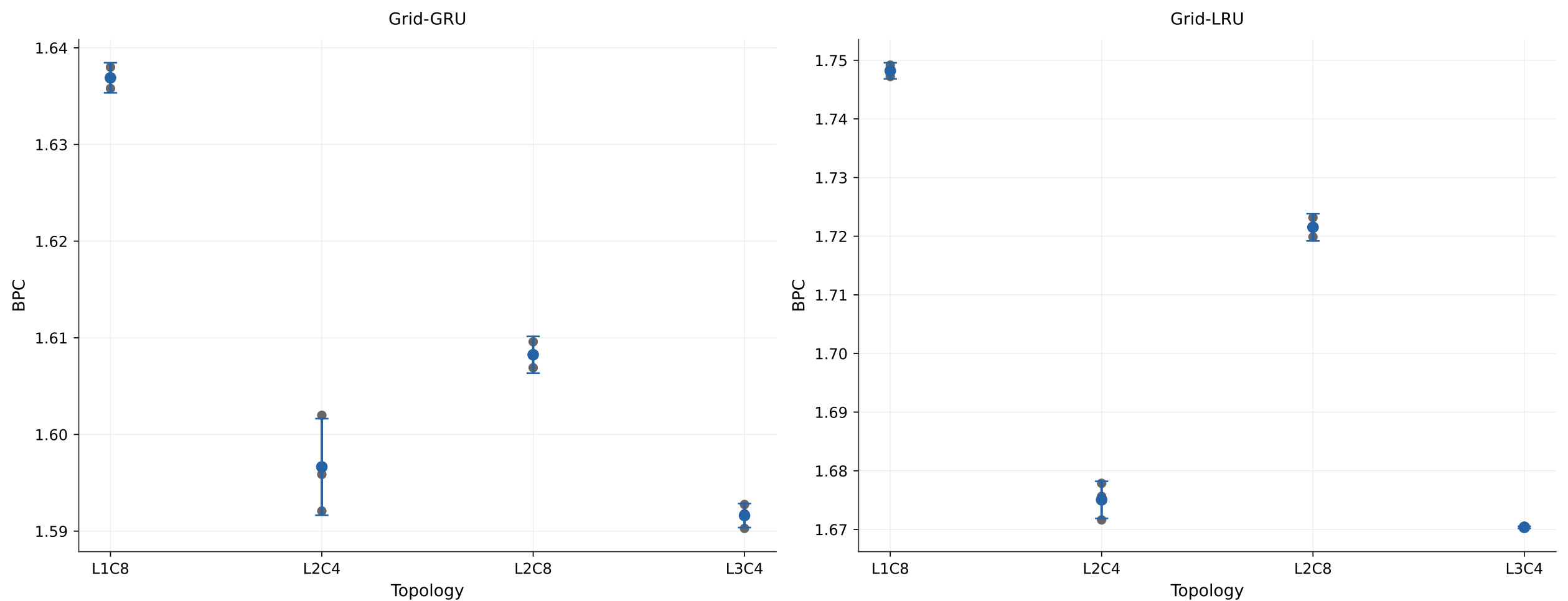
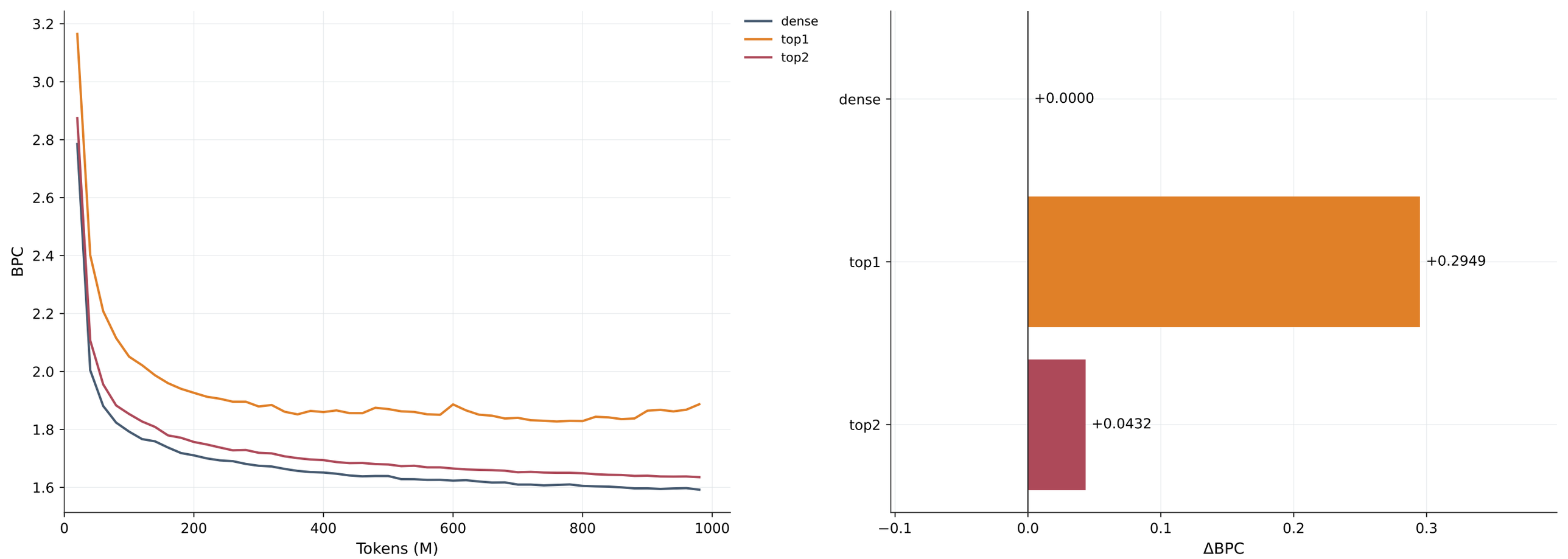
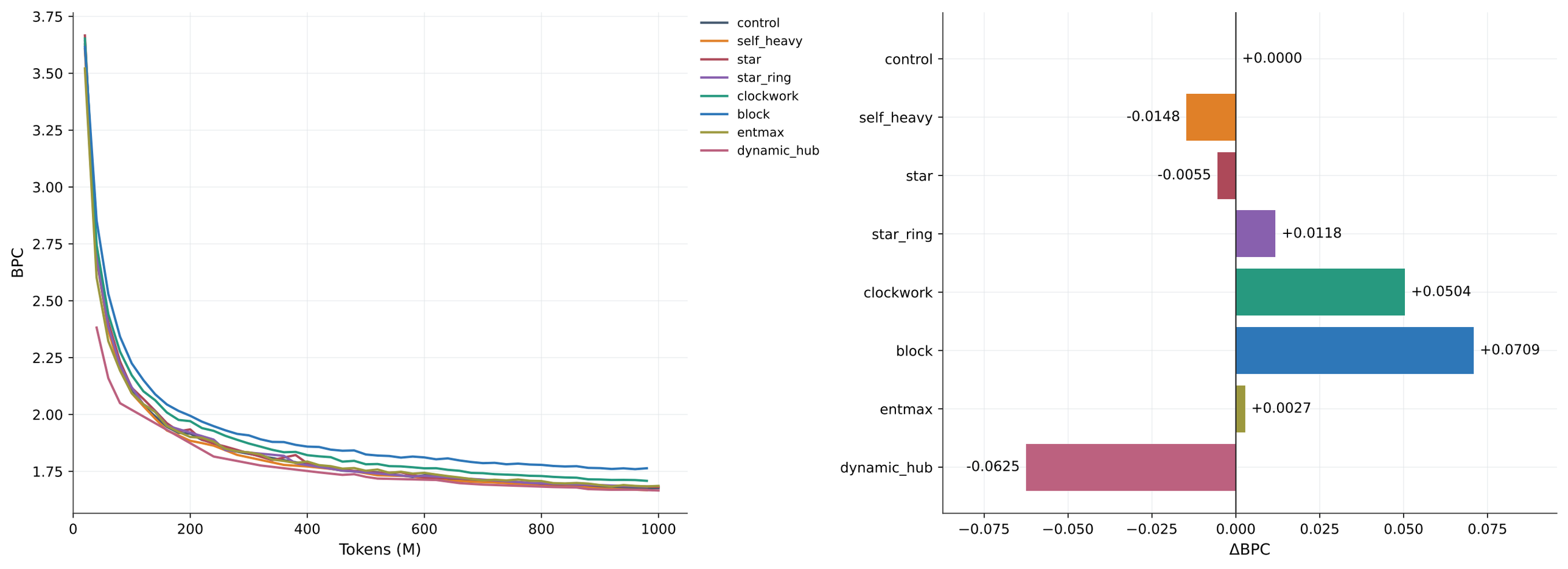

Mamba (n=2)
Grid-GRU L3C4 (n=3)
Grid-GRU L2C4 (n=3)
Grid-GRU L2C8 (n=2)
GRU L3 (n=2)
GRU L2 (n=3)
Grid-GRU L1C8 (n=2)
Grid-LRU L3C4 (n=3)
HGRN2 (n=2)
GRU L1 (n=2)
Grid-LRU L2C8 (n=2)
mLSTM (n=2)
Grid-LRU L1C8 (n=2)
DeltaNet (n=2)
RIMs (n=2)
BRIMs (n=2)
1.5651 ± 0.0012
1.5916 ± 0.0012
1.5966 ± 0.0050
1.6082 ± 0.0019
1.6205 ± 0.0076
1.6368 ± 0.0018
1.6369 ± 0.0016
1.6690 ± 0.0052
1.6695 ± 0.0005
1.6807 ± 0.0003
1.7156 ± 0.0047
1.7163 ± 0.0039
1.7473 ± 0.0005
1.7849 ± 0.0005
1.8744 ± 0.0977
1.8968 ± 0.0318
1.6
1.7
1.8
1.9
2.0
BPC

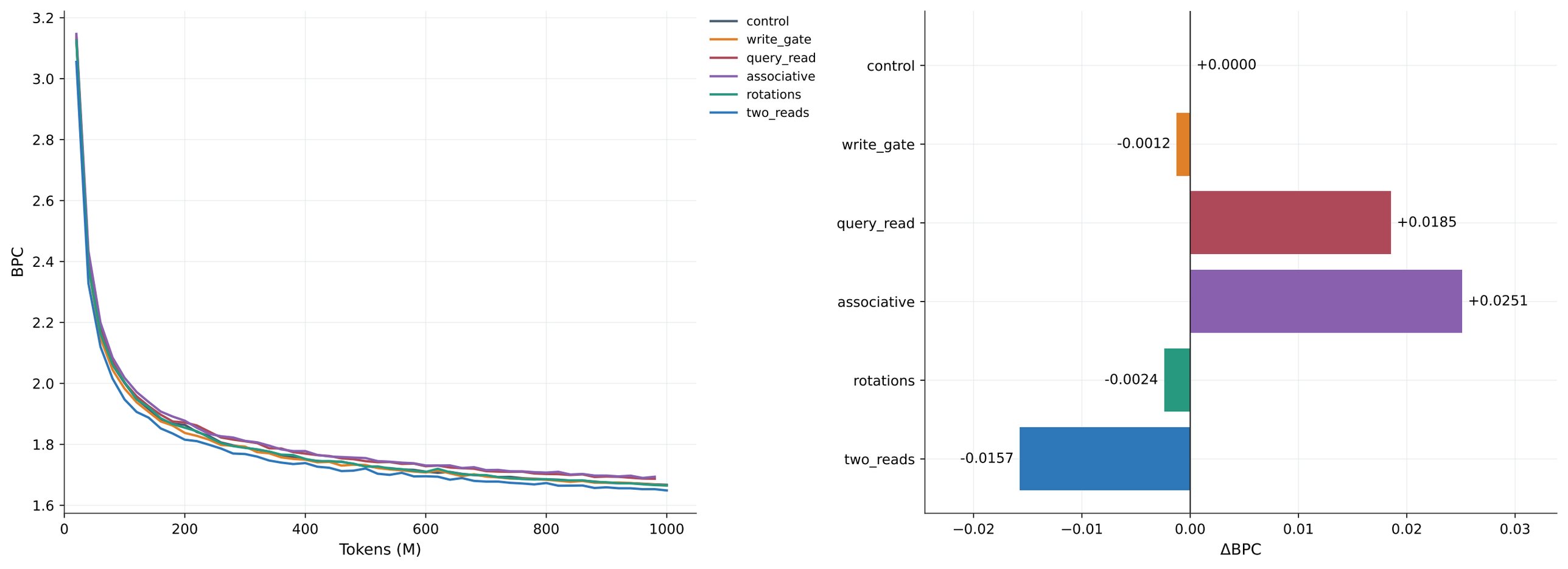

Mamba (n=2)
Grid-GRU L3C4 (n=3)
Grid-GRU L2C4 (n=3)
Grid-GRU L2C8 (n=2)
GRU L3 (n=2)
GRU L2 (n=3)
Grid-GRU L1C8 (n=2)
GRU L1 (n=2)
HGRN2 (n=2)
Grid-LRU L2C4 (n=3)
Grid-LRU L3C4 (n=3)
mLSTM (n=2)
Grid-LRU L2C8 (n=2)
Grid-LRU L1C8 (n=2)
DeltaNet (n=2)
RIMs (n=2)
BRIMs (n=2)
1.6158
1.6475
1.6508
1.6614
1.6715
1.6813
1.6912
1.7198
1.7348
1.7719
1.7737
1.7763
1.8269
1.8470
1.8577
1.9908
2.0757
1.6
1.7
1.8
1.9
2.0
2.1
BPC

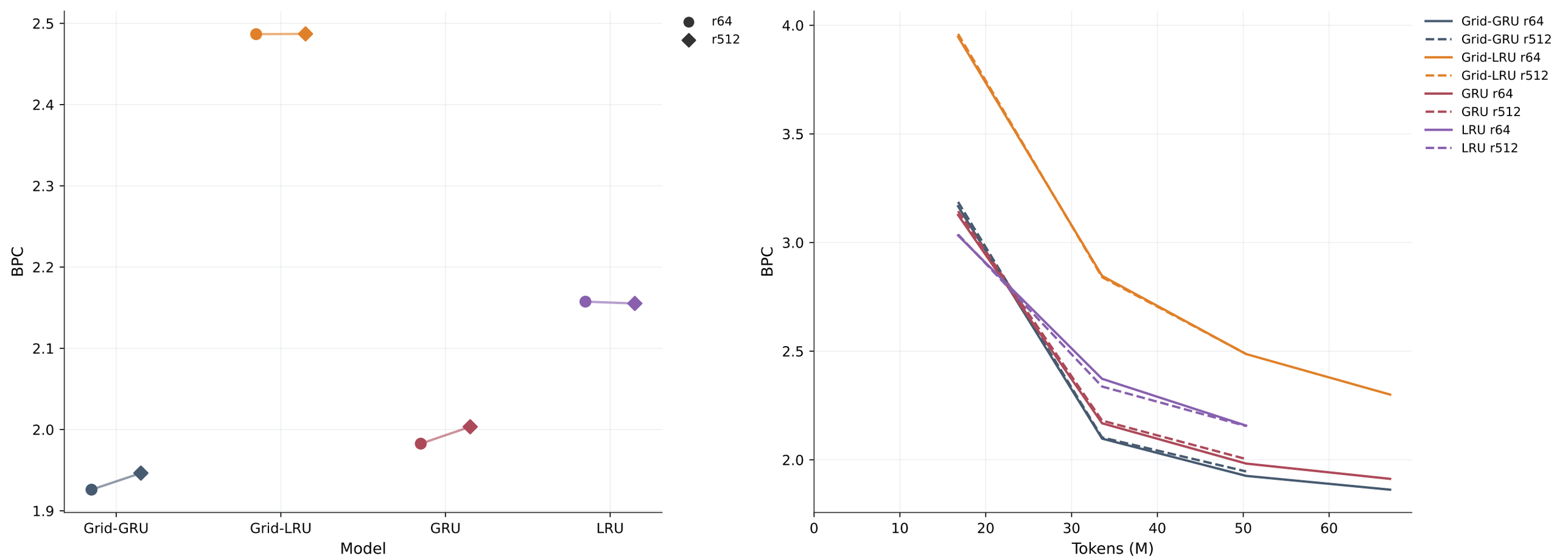

Transformer
BPC
Tokens (M)
2.4
2.2
2.0
1.8
1.6
0
200
400
600
800
1000
Grid-GRU L2C4
GRU L3
Mamba
DeltaNet
HGRN2
mLSTM
RIMs
BRIMs
Transformer seed 42
Transformer seed 1337

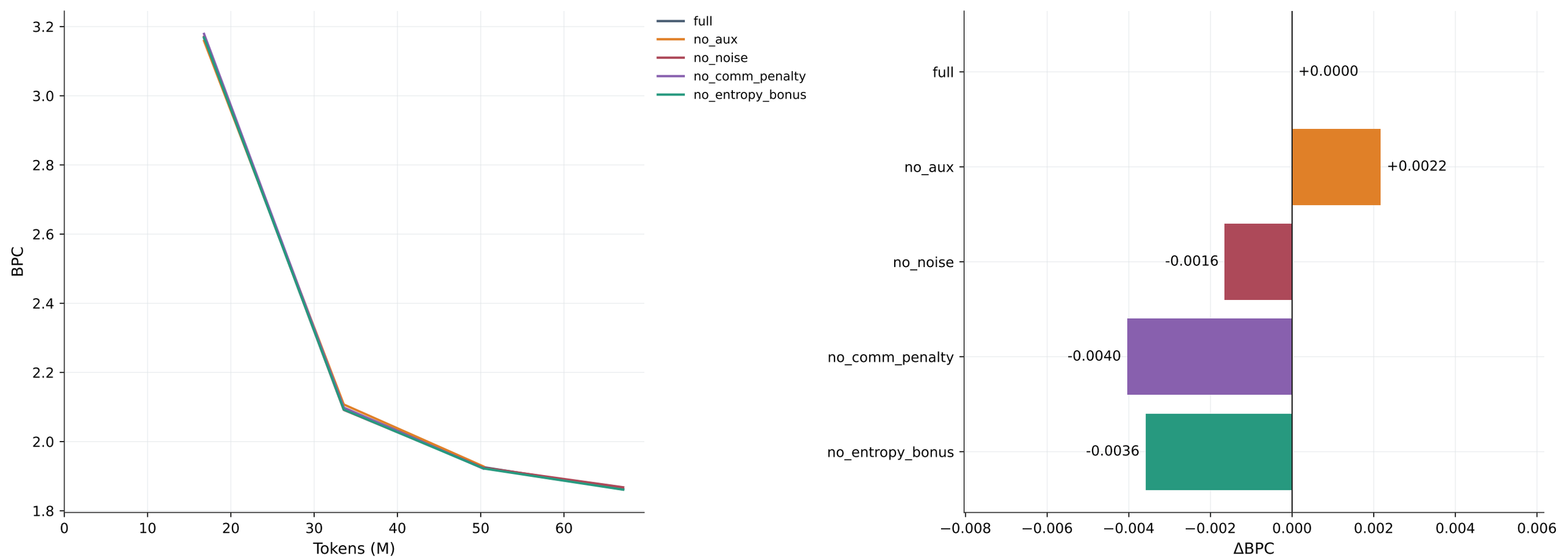

Grid-GRU
BPC
1.64
1.63
1.62
1.61
1.60
1.59
L1C8
L2C4
L2C8
L3C4
Topology
Grid-LRU
BPC
1.75
1.74
1.73
1.72
1.71
1.70
1.69
1.68
1.67
L1C8
L2C4
L2C8
L3C4
Topology

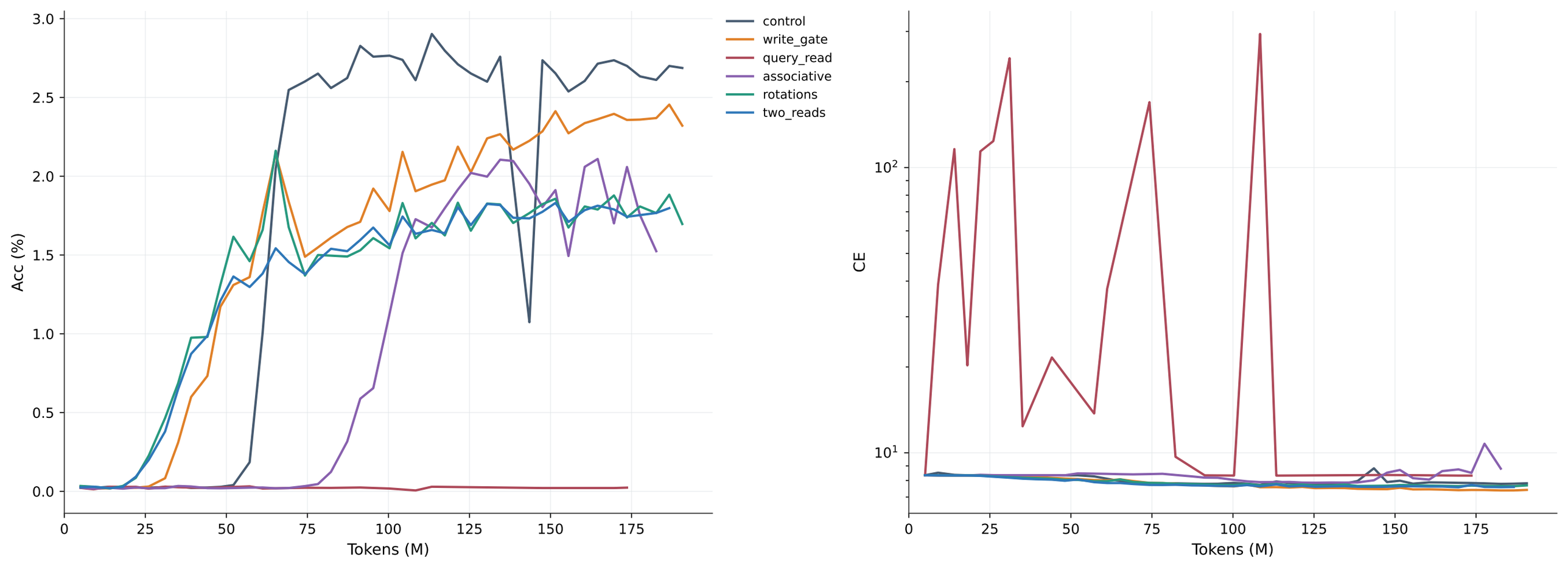

dense
top1
top2
BPC
3.2
3.0
2.8
2.6
2.4
2.2
2.0
1.8
1.6
0
200
400
600
800
1000
Tokens (M)
dense
top1
top2
+0.0000
+0.2949
+0.0432
−0.1
0.0
0.1
0.2
0.3
ΔBPC

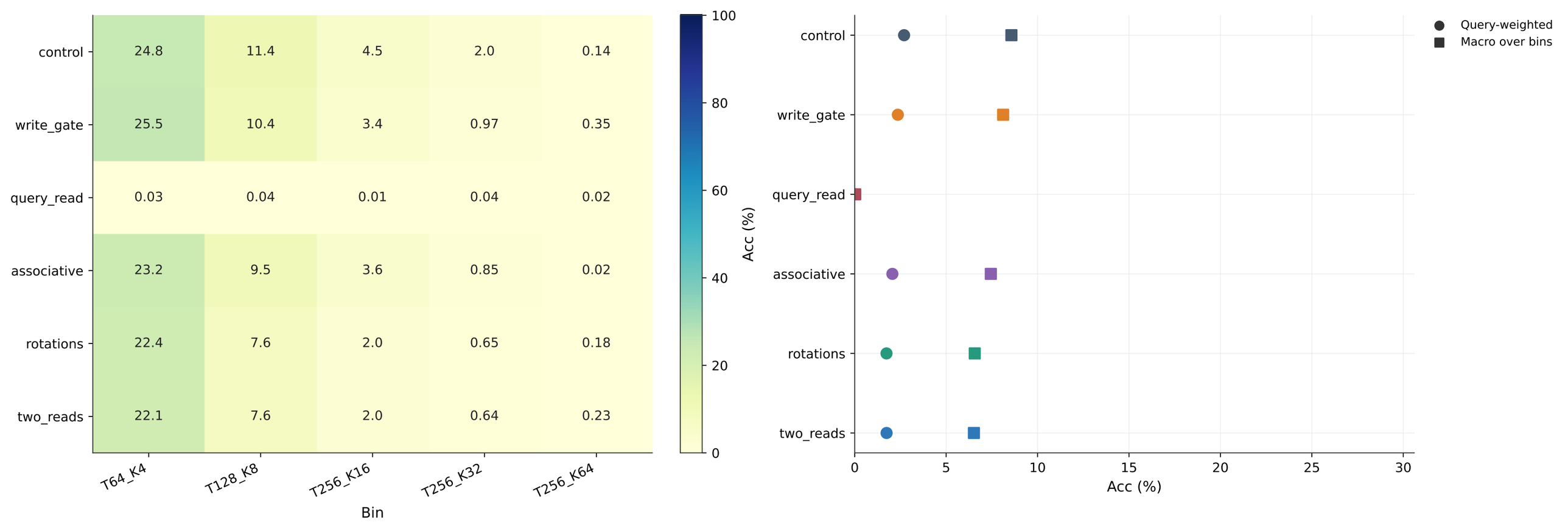

BPC
Tokens (M)
control
self_heavy
star
star_ring
clockwork
block
entmax
dynamic_hub
+0.0000
-0.0148
-0.0055
+0.0118
+0.0504
+0.0709
+0.0027
-0.0625
ΔBPC

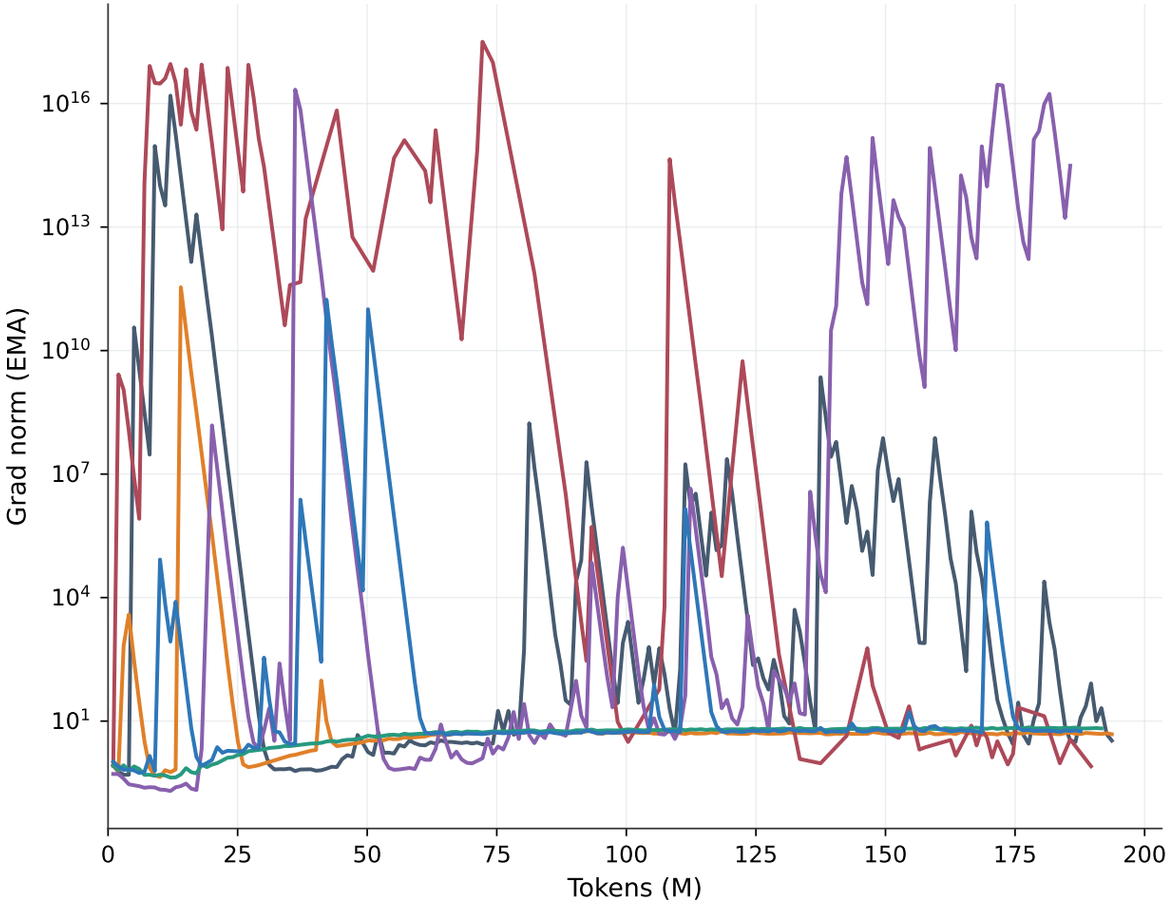

BPC
Tokens (M)
control
write_gate
query_read
associative
rotations
two_reads
+0.0000
-0.0012
+0.0185
+0.0251
-0.0024
-0.0157
ΔBPC

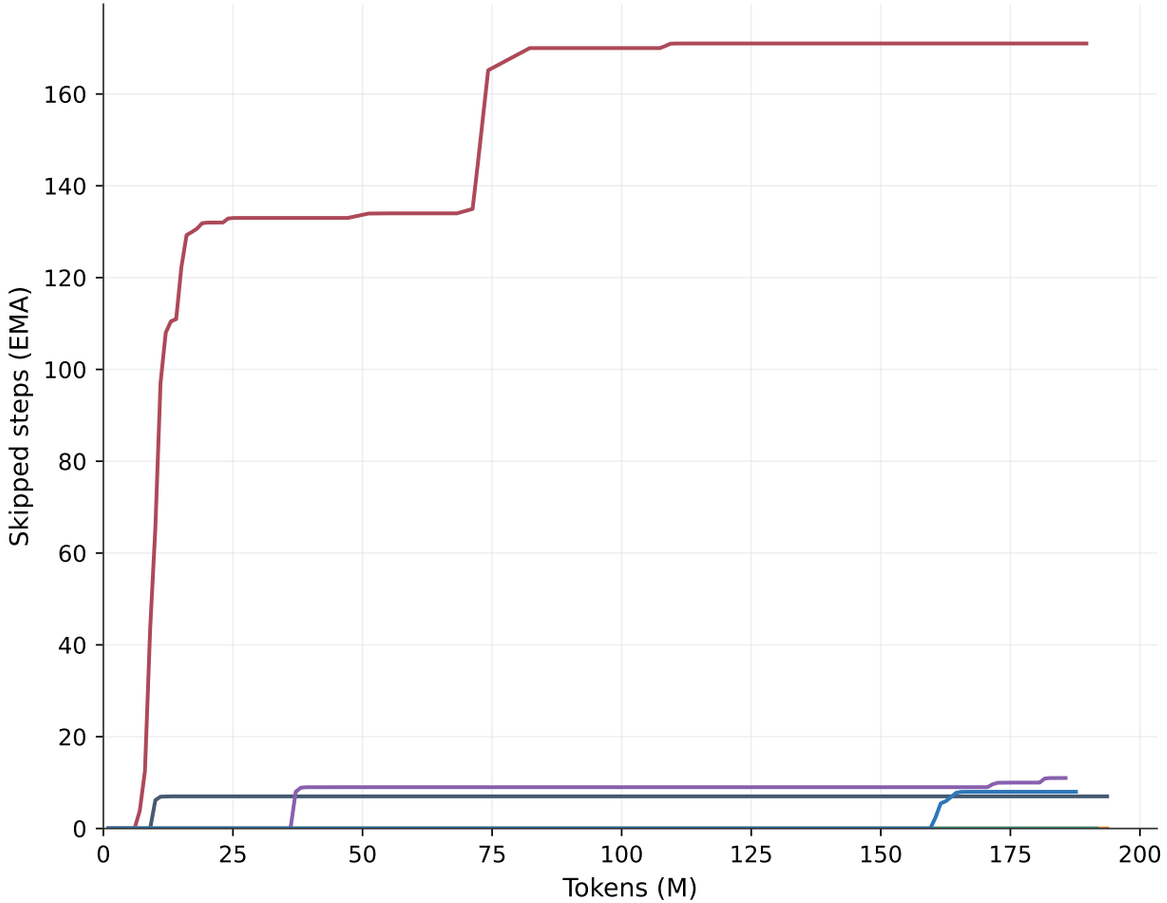

BPC
2.5
2.4
2.3
2.2
2.1
2.0
1.9
Grid-GRU
Grid-LRU
GRU
LRU
Model
r64
r512
BPC
4.0
3.5
3.0
2.5
2.0
0
10
20
30
40
50
60
Tokens (M)
Grid-GRU r64
Grid-GRU r512
Grid-LRU r64
Grid-LRU r512
GRU r64
GRU r512
LRU r64
LRU r512

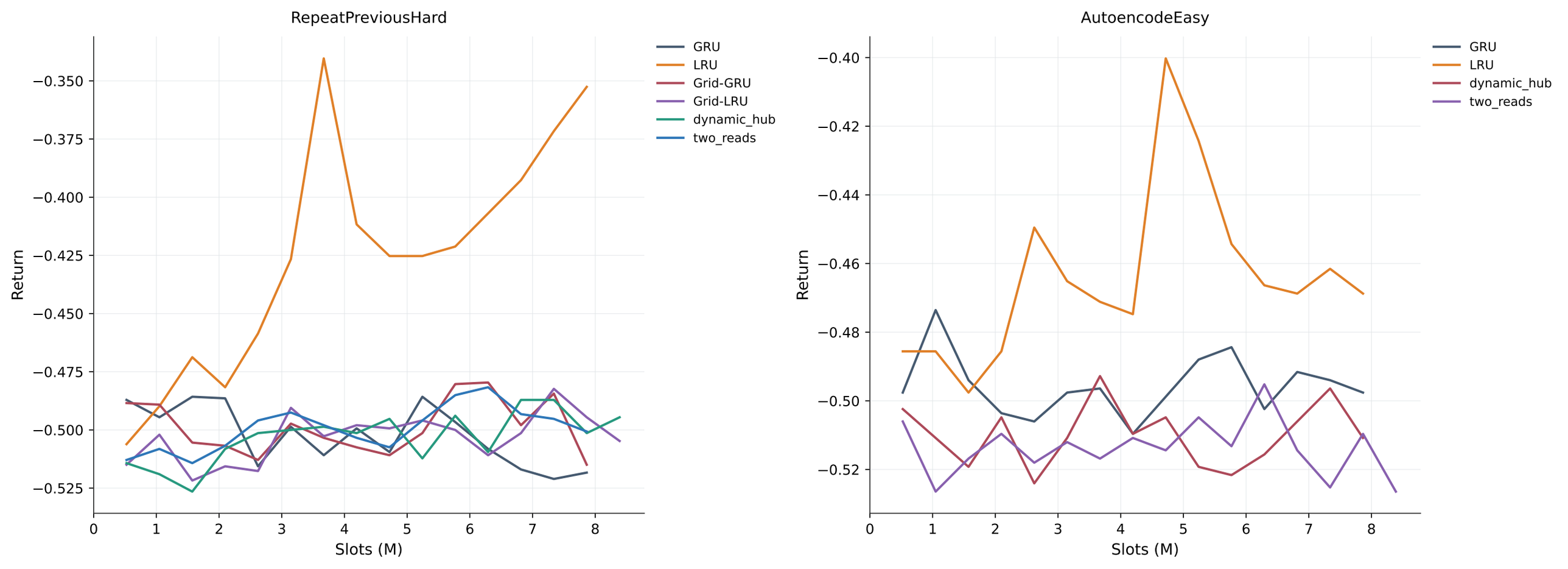

full
no_aux
no_noise
no_comm_penalty
no_entropy_bonus
BPC
3.2
3.0
2.8
2.6
2.4
2.2
2.0
1.8
0
10
20
30
40
50
60
Tokens (M)
full
no_aux
no_noise
no_comm_penalty
no_entropy_bonus
+0.0000
+0.0022
-0.0016
-0.0040
-0.0036
−0.008
−0.006
−0.004
−0.002
0.000
0.002
0.004
0.006
ΔBPC

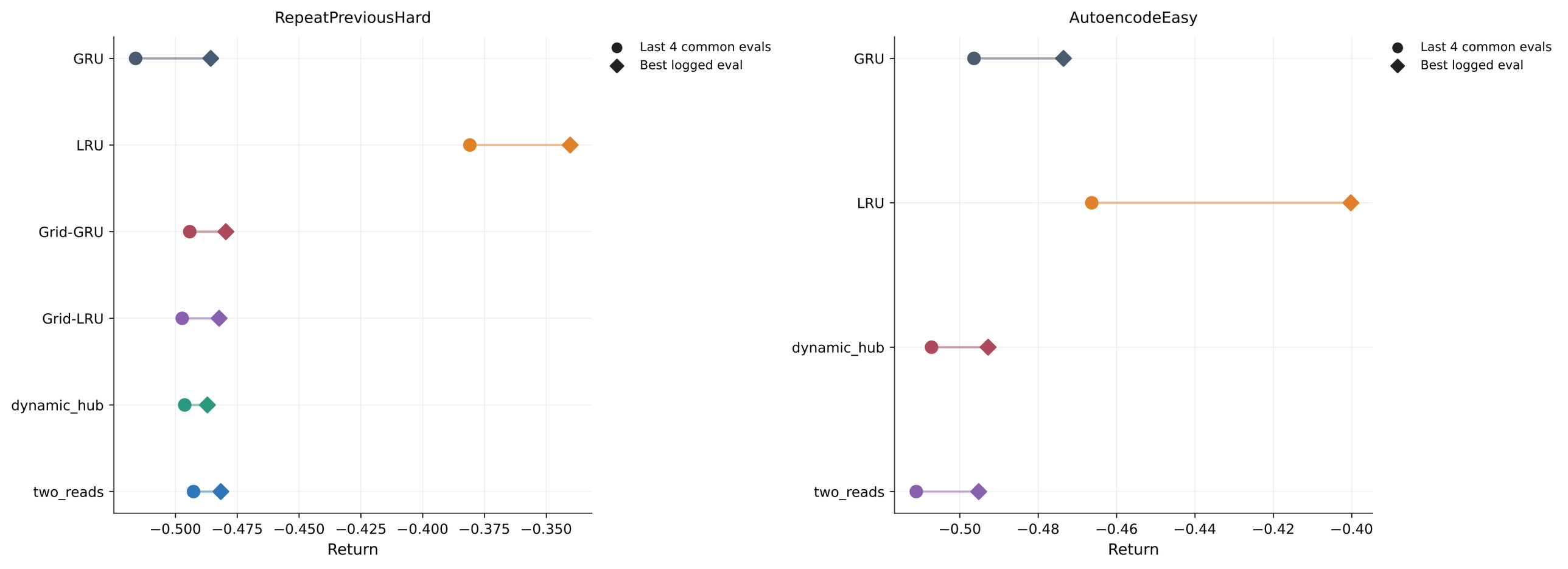

control
write_gate
query_read
associative
rotations
two_reads
Acc (%)
Tokens (M)
CE
Tokens (M)


control
write_gate
query_read
associative
rotations
two_reads
24.8
11.4
4.5
2.0
0.14
25.5
10.4
3.4
0.97
0.35
0.03
0.04
0.01
0.04
0.02
23.2
9.5
3.6
0.85
0.02
22.4
7.6
2.0
0.65
0.18
22.1
7.6
2.0
0.64
0.23
T64_K4
T128_K8
T256_K16
T256_K32
T256_K64
Bin
100
80
60
40
20
0
Acc (%)
Query-weighted
Macro over bins
0
5
10
15
20
25
30
Acc (%)


Grad norm (EMA)
10^{16}
10^{13}
10^{10}
10^{7}
10^{4}
10^{1}
0
25
50
75
100
125
150
175
200
Tokens (M)
control
write_gate
query_read
associative
rotations
two_reads


Skipped steps (EMA)
0
20
40
60
80
100
120
140
160
0
25
50
75
100
125
150
175
200
Tokens (M)


RepeatPreviousHard
Return
−0.350
−0.375
−0.400
−0.425
−0.450
−0.475
−0.500
−0.525
0
1
2
3
4
5
6
7
8
Slots (M)
GRU
LRU
Grid-GRU
Grid-LRU
dynamic_hub
two_reads
AutoencodeEasy
Return
−0.40
−0.42
−0.44
−0.46
−0.48
−0.50
−0.52
0
1
2
3
4
5
6
7
8
Slots (M)
GRU
LRU
dynamic_hub
two_reads


RepeatPreviousHard
GRU
LRU
Grid-GRU
Grid-LRU
dynamic_hub
two_reads
−0.500
−0.475
−0.450
−0.425
−0.400
−0.375
−0.350
Return
Last 4 common evals
Best logged eval
AutoencodeEasy
GRU
LRU
dynamic_hub
two_reads
−0.50
−0.48
−0.46
−0.44
−0.42
−0.40
Return
Last 4 common evals
Best logged eval